Jose Lobo

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PROFESSIONAL SUMMARY

Data Scientist with 4+ years of experience delivering end-to-end analytics and machine learning solutions in healthcare. Proven ability to perform data preparation, exploratory data analysis (EDA), predictive modeling, and dashboard development to support business and clinical decision-making. Strong communicator with deep expertise in SQL, Python, R, and cloud data platforms, plus a background in biology and four years of medical school. Hands-on with agentic coding, cloud and local large language models, RAG and LangChain, and building self-hosted AI systems and data pipelines.

PROFESSIONAL EXPERIENCE

Envision Healthcare · Data Scientist November 2025 – Present

- Lead end-to-end data science workflows including data ingestion, cleaning, EDA, feature engineering, model development, and evaluation.
- Develop and evaluate machine learning models (classification and regression) using Python and R (scikit-learn, XGBoost) to support revenue recovery initiatives.
- Partner with the team to turn denied-claim classification models into products that identify reversible denials and recover revenue from health insurance providers.
- Build forecasting models that predict emergency department patient volumes, so operations teams can plan staffing and allocate resources ahead of demand.
- Design and maintain executive-ready dashboards and recurring reports using Tableau, Power BI, R Shiny, and Streamlit.
- Translate complex analytical findings into actionable insights for technical and non-technical stakeholders.
- Document data definitions, modeling assumptions, and methodologies to ensure reproducibility and audit readiness.

Independent AI Engineering · Personal projects Ongoing

- Develop software with agentic coding workflows and agent harnesses, directing AI agents to plan, write, test and refactor code, including this interactive portfolio.
- Work fluently across cloud and local large language models, including Claude, OpenAI, DeepSeek, MiniMax and MiMo, choosing the right model for cost, privacy and quality.
- Build retrieval-augmented generation (RAG) applications with LangChain that ground model answers in curated document collections.
- Design data pipelines that collect and clean public datasets, then train, evaluate and fine-tune models on personal workstations.
- Assemble and maintain personal AI ecosystems from open-source tools, managing multiple systems across my own computers.
- Apply Laya classification models in projects, and understand how agents and models are built, trained and fine-tuned end to end.

Envision Healthcare · Business Analyst II (Data Scientist Responsibilities) December 2022 – November 2025

Although the formal Data Scientist title did not exist until November 2025, this role performed core data science functions aligned to that title.

- Collected, cleaned, and prepared large-scale structured and semi-structured healthcare data from Amazon Redshift (AWS) and Oracle.
- Performed exploratory data analysis to identify trends, anomalies, and data quality issues impacting financial and clinical outcomes.
- Designed, trained, and evaluated machine learning models to predict and reverse denied medical claims.
- Built ETL pipelines, analytical datasets, and feature tables to support modeling and downstream analytics.
- Delivered dashboards and analytical applications using Tableau, Power BI, R Shiny, and Streamlit, including reporting for leadership and litigation.
- Optimized complex SQL queries and refactored data sources to improve performance and reliability and reduce duplication.
- Served as a Tableau Server administrator, helping users with access and troubleshooting of reports and dashboards.
- Validated and troubleshot data discrepancies with SQL in DBeaver; managed work and reported status to leadership through JIRA.

HCA Healthcare · Data Analyst April 2022 – December 2022

- Developed clinical and operational dashboards using Power BI and Tableau alongside Excel, Teradata, and SharePoint.
- Served as subject matter expert and first-line support for clinicians using enterprise analytics products.
- Resolved production data issues through debugging, root-cause analysis, and quality assurance testing.
- Validated data discrepancies with SQL in Teradata and helped implement and manage report security.
- Created product documentation, including requirements and data guides.
- Trained and supported facility-level users by planning and leading tool demonstrations, interpreting dashboard data, and assisting in opportunity discovery.

DCIDS · Process Technician IV July 2019 – June 2021

- Processed and packaged human tissue for a tissue bank in a sterile surgical setting, handling surgical instruments, power and laboratory equipment, chemical reagents, and biohazardous waste.
- Prepared processing and packaging records, including graft and anatomical identification, measurements, and data entry.
- Performed clean room sterilization, testing, and environmental monitoring; assisted with inventory, research manipulations, and training new hires.
- Collaborated across departments to ensure allograft orders were met on time.

Amazon · Picker, Packer, Inventory Control Quality Assurance October 2017 – September 2018

- Retrieved and quality-checked items for order fulfillment and supported inventory and quality control as part of a nightly-goal team.

Kroger · Produce Overnight Associate October 2017 – February 2018

- Reconciled piece counts against invoices, rotated and stocked produce, managed markdowns and waste, and served customers. Trained in Powered Industrial Trucks (PIT).

TECHNICAL SKILLS

Programming & Analytics: Python, SQL, R, JavaScript (incl. D3.js), C++, HTML, CSS, JSON; Machine Learning, Predictive Modeling, EDA, Feature Engineering

AI & Agentic Engineering: Agentic coding and agent harnesses; cloud and local LLMs (Claude, OpenAI, DeepSeek, MiniMax, MiMo); retrieval-augmented generation (RAG); LangChain; model fine-tuning; Laya classification models; open-source AI tooling and self-hosted AI systems

Data Platforms: Amazon Redshift (AWS), Oracle, Teradata, Microsoft SQL Server, MySQL, SQLite, MongoDB, DBeaver

BI & Visualization: Tableau (incl. Server administration), Power BI (DAX), R Shiny, Streamlit, Qlik Sense, QlikView, Excel

Data Engineering: ETL pipelines, analytical datasets and feature tables

Tools: GitHub, JIRA, Agile, Jupyter, VS Code, RStudio, SharePoint, ServiceNow (Service Central), command line, Windows, macOS

Other: EMR software, laboratory skills (QC, light microscopy, gel electrophoresis, equipment calibration), fluent in Spanish and English

EDUCATION

Bachelor of Science in Biology (Cum Laude) · Middle Tennessee State University, Murfreesboro, TN · Concentration: Physiology · Minor: Chemistry

Medical school coursework and clerkships (four years) · University of Medicine and Health Sciences, St. Kitts

Computer science (self-directed) · freeCodeCamp.org and Udemy

CERTIFICATIONS

- Machine Learning with Python (freeCodeCamp), October 2021
- Data Analysis with Python (freeCodeCamp), October 2021
- Scientific Computing with Python (freeCodeCamp), October 2021
- Data Visualization (freeCodeCamp), September 2021
- Front End Development Libraries (freeCodeCamp), September 2021
- Back End Development and APIs (freeCodeCamp), September 2021
- JavaScript Algorithms and Data Structures (freeCodeCamp), August 2021
- Responsive Web Design (freeCodeCamp), July 2021
- Basic Life Support (BLS) and Advanced Cardiovascular Life Support (ACLS), January 2015
- HIPAA, Infection Control, Needlestick Prevention and Bloodborne Pathogens training (OSHA standards), May 2014

CLINICAL EXPERIENCE

- 2016–2017: Medical student clerkships in psychiatry, general surgery, emergency medicine, pain management and orthopedic surgery, DeKalb Medical, Atlanta, GA.
- 2015: Medical student clerkship in internal medicine, Weiss Memorial Hospital, Chicago, IL.
- 2014–2015: Medical student clerkships in OB/GYN, pediatrics, family medicine, emergency medicine and orthopedic surgery, DeKalb Medical Center, Atlanta, GA.
- 2013–2014: Medical student clerkship in cardiology, St. Mary's Regional Medical Center, Lewiston, ME.
- Summer 2010: Premed Internship Program, Jackson-Madison County General Hospital, Jackson, TN (254 hours shadowing surgeons, physicians and nurses).

VOLUNTEER EXPERIENCE

- 2015: Stop Hunger Now, gathering food for people around the world.
- 2010: Food server, Regional Inter-Faith Association (RIFA) Soup Kitchen, Jackson, TN.
- 2008–2009: Push America, building and fundraising for accessible environments for people with special needs (Murfreesboro, TN and Hattiesburg, MS).
- Fall 2008: Rivalry Run 5K fundraiser for people with special needs.

PRESENTATIONS AND LECTURES

- January 2017: Pathology, clinical diagnosis and treatment of Parsonage-Turner syndrome.
- March 2016: Embryology, anatomy and pathology of the thyroid gland.
- January 2016: Differential diagnoses for myositis in HIV patients; diagnosing infectious endocarditis with the Duke criteria.
- December 2015: Sick sinus syndrome in a clinical setting; infectious causes of adrenal insufficiency.
- October 2015: Platelet-rich plasma and its treatment applications for sports injury.
- May 2015: Differential diagnoses of hypertension during pregnancy; pathology of leiomyomas.